

Multiple Choice (10 marks)

1. Statement 1 | If H is a subgroup of a group G and a belongs to G , then $aH = Ha$.
Statement 2 | If H is normal of G and a belongs to G , then $ah = ha$ for all h in H .
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. Compute the product in the given ring. $(12)(16)$ in $\mathbb{Z}_{24}$
 - (a) 0
 - (b) 1
 - (c) 4
 - (d) 6
3. For $T: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ where $T(1, 0) = 3$ and $T(0, 1) = -5$, find $T(-3, 2)$.
 - (a) -19
 - (b) -10
 - (c) 19
 - (d) 10
4. Find the characteristic of the ring $\mathbb{Z}_3 \times \mathbb{Z}_3$.
 - (a) 0
 - (b) 3
 - (c) 12
 - (d) 30
5. Statement 1 | Any set of two vectors in $\mathbb{R}^2$ is linearly independent. Statement 2 | If $V = \text{span}(v_1, \dots, v_k)$ and $\{v_1, \dots, v_k\}$ are linearly independent, then $\dim(V) = k$.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

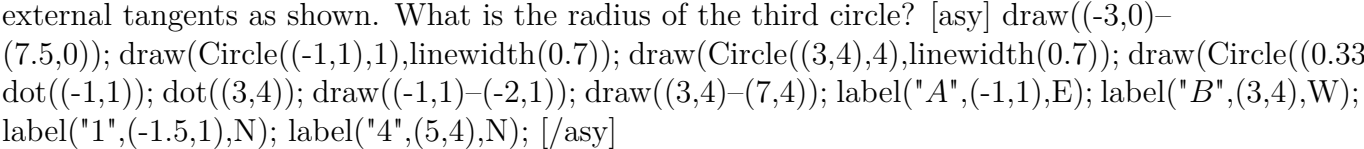
True / False (10 marks)

Answer True or False

6. True or False: In the finite field $\mathbb{Z}_7$, the element 2 is a generator of the multiplicative group of nonzero elements.
7. True or False: If $3x + 7y$ is divisible by 11, then $9x + 4y$ must also be divisible by 11 for positive integers x and y .
8. True or False: In the group $G = \{2, 4, 6, 8\}$ under multiplication modulo 10, the identity element is 6.
9. True or False: It is possible to have a nontrivial homomorphism from a finite group into an infinite group.
10. True or False: If a and b are elements of finite order in an Abelian group, then $|ab|$ is the least common multiple of $|a|$ and $|b|$ under certain conditions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A circle centered at A with a radius of 1 and a circle centered at B with a radius of 4 are externally tangent. A third circle is tangent to the first two and to one of their common external tangents as shown. What is the radius of the third circle? 
12. Discuss the conditions under which a triangular region, with one vertex at the center of a circle of radius 1 and the other two vertices on the circle, achieves its maximum area. Calculate this area and justify your reasoning.

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